

When Should a Vision Specialist Defer?

Budgeted Collaboration with Vision–Language Models

Supplementary Material

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This supplement is independently compilable. Sections A–F document the final common routing recipe and distinguish it from earlier experiments. Section G adds archived label-volume and joint-training studies; Sections H–J provide cases, cost records and artifact provenance. M and Q denote MiniCPM-V-4.5 and Qwen3.8-27B-FP8. No additional model training or inference was performed to prepare this supplement.

A Data Preparation and Evaluation

Table S1 lists the examples actually used in this study. The four datasets span detection-related localization and event recognition, together with classification and visual reasoning. They are not four object-detection benchmarks, and their task scores have different meanings.

Table S1: Experimental splits. Counts refer to task examples, not uniformly to distinct images. Training denotes the router’s full available training split; specialist training provenance is listed separately. Totals are the sums of the three columns after the recorded exclusions.

Dataset	Example unit	Train	Validation	Test	Total	Evaluation
CUB-200-2011	One image	4,792	1,200	5,794	11,786	Top-1 accuracy
gRefCOCO	Image–expression pair	209,344	14,229	35,263	258,836	Strict localization accuracy
NLVR2	Two images and a statement	85,042	6,982	13,937	105,961	Binary accuracy
ConstructionSite10k	One image, four event labels	6,308	701	3,004	10,013	Event macro-F1

CUB-200-2011 [1] tests fine-grained recognition among 200 bird species, where differences in plumage, shape and local appearance distinguish related categories. We report top-1 class accuracy. The router reserves 1,200 images from the official training partition for validation and excludes two training images identified as near-duplicates of test images, leaving 4,792 training examples. The official 5,794-image test partition is retained.

gRefCOCO [2] tests generalized referring-expression localization, including expressions with one, multiple or no matching objects. TestA and testB contribute 19,200 and 16,063 expressions. Our fixed box-based evaluation requires a valid empty answer for no-target examples. Boxes from either model are first filtered at score ≥ 0.7 . For target-present examples, greedy one-to-one matching uses generalized intersection over union of at least 0.5; an example is correct only when instance F1 equals one, with no missed or extra instances. This strict correctness measure is distinct from the dataset’s official segmentation evaluation and from detection average precision.

NLVR2 [3] tests whether a natural-language statement is true of an ordered pair of natural images, including counting, attributes and relations across the pair. We report binary accuracy. The prepared training set contains 85,042 usable examples after filtering the source training partition; validation has 6,982, and the two test portions have 6,967 and 6,970. Routing acts on the complete two-image example.

ConstructionSite10k [4] contains construction scenes with structured safety annotations. We evaluate four event-presence categories: personal-protection, fall-protection, unprotected-edge and excavator-proximity violations. The original 7,009 training images are split into 6,308 for training and 701 for validation; all 3,004 test images are retained. The dataset name is approximate: the prepared partitions contain 10,013 images. We report

$$M_{\text{event}} = \frac{1}{4} \sum_{j=1}^4 \frac{2TP_j}{2TP_j + FP_j + FN_j}, \quad (\text{S1})$$

with zero for a class whose denominator is zero. Delegation replaces all four event predictions for an image; confusion counts are then recomputed. This is event recognition, not bounding-box average precision.

A.1 Examples from the Four Datasets

CUB-200-2011



Nashville Warbler (class 172)

gRefCOCO



Query: a mouse beside a laptop and a little boy sitting far from us is looking at a laptop.
Targets: 2.

NLVR2



Statement: The right image shows a curving walkway of dark glass circles embedded in dirt and flanked by foliage. Label: False.

ConstructionSite



Four safety-event labels: all absent.

Figure S1: Real validation examples and their recorded annotations. CUB and ConstructionSite use one image, NLVR2 uses an ordered image pair, and gRefCOCO couples an image with an expression. The first record in each validation manifest was selected before viewing the images. Images illustrate the tasks rather than model performance.

The gRefCOCO example refers to two separate targets, illustrating why all matching objects must be found. The NLVR2 label evaluates the full statement about the ordered pair. For ConstructionSite, an all-negative image still requires evaluating all four event categories. The sample identifiers and original annotations are provided with the figure data.

A.2 Processing Order and Unit of Analysis

The router operates on prepared task examples, not on a universal image count. Source partitions and recorded exclusions are applied before joining paired model predictions, task labels and features by sample identity. Multiple referring expressions may share an image; an NLVR2 example retains its ordered image pair. These repeated-image relationships should also be

retained when constructing uncertainty estimates. No uncertainty interval is inferred from treating all expressions as independent photographs.

For gRefCOCO, the score threshold precedes matching for both endpoints. A target-present prediction is correct only if every ground-truth instance is matched and no extra instance remains. For no-target expressions, a valid empty prediction is required; an invalid response is not a correct empty answer. The generalized IoU threshold is 0.5, not ordinary IoU. This measure evaluates strict expression-level correctness and must not be relabeled as official segmentation accuracy.

The construction labels are four event-presence outputs, not a detector’s average precision. An image may contain events in different paired-outcome states. The router makes one decision for the image and replaces all four outputs together. Invalid responses are recorded as invalid in the prediction interface. For the archived macro-F1 calculation, only literal positive predictions count as positives: invalid outputs can cause false negatives on positive events without creating false positives on negative events. Paired-outcome correctness and macro-F1 therefore have different accounting, and the tables retain the archived convention.

B Specialists, Distillation and Router Training

Specialists and general models. We pair ResNet-50 [5] and GLSim [6] on CUB, GroundingDINO [7] and InstanceVG [8] on gRefCOCO, BEiT3 [9] and ViLT [10] on NLVR2. Construction pairs YOLO26x [11] and RT-DETR-X [12]. Each specialist is paired with MiniCPM-V-4.5 [13] and Qwen3.8-27B-FP8 [14], abbreviated M and Q. “Small” identifies the task specialist’s role; it does not denote a common parameter budget across specialists.

Table S2: Specialist provenance and training performed in this project. Student distillation and router fitting are described in the text. Reused checkpoints are not presented as models trained by this study.

Specialist	Initialization / status	Recorded training or inference settings
ResNet-50	PyTorch pretrained weights; 200-class head trained here	Original 5,994 CUB training images; 5 epochs, batch 64, AdamW 3×10^{-4} , weight decay 0.05, cosine decay, label smoothing 0.1. Random 224^2 crops and flips for training; resize 256 and center crop 224 for evaluation. The internal router-validation images were included.
GLSim	Author’s CUB checkpoint	Global/local model with ViT-B/16; resize 300 then center crop 224; no test-time flip. Original training provenance is not independently confirmed.
GroundingDINO	Pretrained tiny checkpoint	Reused O365/GoldG/V3Det checkpoint; no project training on gRefCOCO. Fixed box threshold 0.7 and text threshold 0.25.
InstanceVG	Author’s gRefCOCO checkpoint	Reused task checkpoint; 320^2 images, text limit 50 tokens, inference batch 16 and score threshold 0.7.
BEiT3	Existing NLVR2 checkpoint	Reused base patch16/224 task checkpoint; no new training. Original third-party fine-tuning logs are not available in the archive.
ViLT	Existing NLVR2 checkpoint	Reused B/32 task checkpoint; ordered two-image input and text limit 40 tokens. Original fine-tuning parameters are not independently audited.
YOLO26x	Pretrained; fine-tuned here	Construction 6,308/701 train/validation; 640^2 input, batch 16, maximum 100 epochs, patience 20, seed 20260831; 92 recorded epochs, mixed precision disabled.
RT-DETR-X	Pretrained; fine-tuned here	Same construction split and training limits; 39 recorded epochs. Both construction runs use automatic optimizer selection; the nominal configured learning rate 0.01 is not treated as the verified effective optimizer rate.

Table S2 separates models trained in this project from reused checkpoints. The construction detectors are converted to event outputs under the recorded task mapping, with fixed detection thresholds of 0.005 for YOLO26x and 0.05 for RT-DETR-X. GroundingDINO uses box and text thresholds of 0.7 and 0.25; InstanceVG uses its recorded 0.7 score threshold. These are fixed specialist operating points, not parameters optimized by the router. Consequently, measured complementarity depends on the underlying specialist configuration as well as on its architecture.

Paired predictions and invalid outputs. Task-specific prompts request a bird class, a binary judgment, localization boxes or the four safety-event outputs. Sampling temperature is zero; maximum generated lengths are 8 tokens for CUB/NLVR2, 512 for gRefCOCO and 768 for construction. Every image’s fixed candidate classes or event categories are supplied in one request. Each model response is parsed once. An invalid response retains its raw text and error reason, yields no valid prediction, and is marked incorrect; no answer is repaired or inferred. For event macro-F1, only literal positive predictions count as positives, so an invalid output can introduce false negatives without adding false positives on negative events. We retain this convention when rebuilding all comparisons. Predictions, labels and features are joined by sample identity.

Distillation. We distill frozen image and text representations from SigLIP2-base-patch16-224 [15] into randomly initialized

students, using training inputs only. After identity-based image exclusion and normalized-text exclusion against validation and test partitions, the distillation pools contain 115,149 distinct images and 154,118 text strings. The image pool comprises 4,792 CUB, 16,994 gRefCOCO, 87,055 NLVR2 and 6,308 construction images. Text comes from allowed training expressions, statements and construction descriptions; bird labels are not converted to training text. The vocabulary is rebuilt from this permitted pool.

Each student has a two-layer Transformer image encoder and a two-layer text encoder, four attention heads, and a feed-forward width four times the hidden width. The selected 8M model has hidden width 380 and feed-forward width 1,520 in both encoders. Images are resized to 224×224 with 16×16 patches. Text uses a 4,096-entry vocabulary, 32-dimensional input embeddings and at most 64 tokens. Mean-pooled representations are projected to 768 dimensions and normalized. Image and text inputs are sampled independently; for normalized teacher and student features v^T, v^S, t^T, t^S , the loss is

$$\mathcal{L}_{\text{distill}} = (1 - \langle v^S, v^T \rangle) + (1 - \langle t^S, t^T \rangle). \quad (\text{S2})$$

There is no class-label loss or image–text relation-matrix loss. AdamW uses batch size 2,048, learning rate 10^{-4} , weight decay 0.01, cosine decay to zero without warmup, and seed 42. Training runs for 300 epochs of 76 steps, with the smaller image pool cycled to match the text pool. Four sizes, approximately 1M, 2M, 4M and 8M parameters, are trained independently using bfloat16. The shared recipe selects the final 300-epoch 8M checkpoint: 4,135,548 image-encoder and 3,936,780 text-encoder parameters. Only the image encoder is active on CUB and construction.

Router fitting and parameter selection. The final router uses the full training splits in Table S1, frozen distilled encoders, 128-dimensional branch projections, specialist states at 90% relative depth, four-state cross-entropy and natural sampling. AdamW uses learning rate 0.003, weight decay 0.01 and batch size 256. Training lasts at most 50 epochs, with early stopping after eight epochs without improvement in validation Score_{25} . All sixteen fits use seed 42. The projections and head add about 0.30–0.58M parameters per specialist; encoder cost remains part of inference even though cached features accelerate training.

The common recipe was chosen by a sequential search over hidden depth, student size, projection width, learning rate, supervision and sampling. Eighteen common settings require 288 pair-specific fits. Checkpoints and settings use the validation ranking objective in Eq. S4; the final operating threshold follows Sec. C. This is a validation-selected recipe, not an exhaustive search or a configuration chosen without validation feedback.

Baselines and data provenance. We compare specialist-only, VLM-only, fixed confidence-threshold routing and learned routing. Confidence uses the negative of either the maximum class probability or the maximum detection confidence, with missing confidence routed first. It chooses its threshold on the same validation 1–99% budget grid, including the same tie rule, as the learned router. No test label determines either threshold.

Strict student distillation and router fitting exclude the held-out partitions, but cannot remove earlier exposure by reused models. In particular, the CUB ResNet was trained on the original 5,994 training images before the internal validation split was made; GLSim’s earlier training provenance is not independently certified. CUB validation performance is therefore diagnostic. Earlier test results had also been inspected before the shared-recipe study: the new selection procedure avoids test-based tuning, but does not constitute a newly blinded evaluation. These limitations apply to the reported interpretation, not just to implementation details.

B.1 Prompts and Generation Settings

The following templates are copied from the full paired-prediction collection used to construct the routing data. Placeholders are replaced by each sample’s expression or statement. CUB supplies all 200 candidate classes in a single request; its full class list is supplied with the code. Both VLMs receive the same task input, but their templates may differ. All requests use temperature zero, with limits of 8 tokens for CUB and NLVR2, 512 for gRefCOCO and 768 for ConstructionSite. NLVR2 sends the left image followed by the right image, with these roles explicitly marked. Qwen disables thinking for CUB and NLVR2; MiniCPM constrains these answers to the class index or True/False. Localization and event requests use JSON schemas. Invalid answers are retained and counted as errors without repair or a second model call.

B.1.1 CUB-200-2011

MiniCPM-V-4.5

Identify the bird species in this image using the fixed CUB-200-2011 class list below. Choose the class whose visible
→ shape, bill, head pattern, wing pattern, body color, and tail best match the bird. Ignore the background and any
→ non-bird objects. Reply with exactly one class number from 1 to 200 and no other text.

{class_names}

Qwen3.8-27B-FP8

Identify the bird species in the image from the fixed CUB-200-2011 reference below. Each class is paired with three visual
→ traits learned only from official training images. Use the species name to establish the likely bird group, then
→ distinguish close candidates by the visible bill, crown and face markings, wing pattern, breast and belly color, tail,
→ and overall form. The traits are clues rather than absolute rules because pose, sex, age, lighting, and occlusion can
→ change what is visible. Choose the strongest overall match, ignore the background, and reply with exactly one class
→ number from 1 to 200 and no other text.

{class_traits}

B.1.2 gRefCOCO

MiniCPM-V-4.5

Your entire reply is one complete JSON line that begins with {"objects": and ends with the final }. A no-match answer is
→ exactly {"objects":[]}. A one-match answer looks like {"objects":[{"bbox_2d":[120,85,640,910],"score":0.92}]}. Several
→ matches use separate complete entries, for example
→ {"objects":[{"bbox_2d":[80,100,300,700],"score":0.93},{"bbox_2d":[560,120,820,740],"score":0.91}]}. Every bbox_2d
→ contains all four xxyy numbers from 0 to 1000 with x1 smaller than x2 and y1 smaller than y2. Never end after only one
→ coordinate; finish every object, the objects list, and the final brace. Return no other text.

Locate all and only the visible objects that satisfy the entire referring expression below. Every stated type, appearance,
→ position, and spatial relation matters. Interpret left and right from the viewer's perspective, and use front, back,
→ nearest, farthest, and ordinal words to distinguish among similar objects. Some expressions intentionally have no
→ matching object, so use the no-match answer whenever the full expression is not visibly satisfied instead of choosing
→ the closest-looking object. Return one tight box for every distinct match without merging objects into a group box.

Referring expression: {expression}

Qwen3.8-27B-FP8

Locate all and only the visible objects that satisfy the entire referring expression below. Every stated object type,
→ appearance, position, and spatial relation matters. Do not return the closest-looking object when part of the
→ expression is missing or contradicted. Some expressions intentionally have no matching object, so an empty list is a
→ normal answer whenever the full expression is not visibly satisfied.

The expression may identify zero, one, or multiple objects. When separate target descriptions are joined by words such as
→ "and", return a separate tight box for every described target rather than one box around the group. When "and" only
→ joins properties of one object, all those properties must hold for that one box. Do not merge neighboring objects into
→ one box.

Coordinates must be xxyy numbers normalized from 0 to 1000, and each box must have a confidence score from 0 to 1. Reply
→ with exactly one complete JSON object and no Markdown or explanation, using this form:
→ {"objects":[{"bbox_2d":[120,85,640,910],"score":0.92}]}. An answer with no match is exactly {"objects":[]}.

Referring expression: {expression}

B.1.3 NLVR2

MiniCPM-V-4.5

The first image is the left image and the second is the right image. Determine whether the sentence below is true for this
→ ordered pair. A sentence is True only when every part of it is supported by the images. Interpret counts and
→ comparisons exactly, distinguish claims about one image from claims about both images together, and handle words such
→ as all, at least, more, fewer, same, different, and no literally. Ignore details that the sentence does not mention.
→ Answer with exactly True or False and nothing else.

Sentence: {sentence}

Qwen3.8-27B-FP8

The first image is the left image and the second is the right image. Decide whether the sentence below is true about the
→ pair. Count each distinct matching object once in the exact image scope named by the sentence before evaluating more,
→ fewer, equal, at least, at most, or exactly. An identifiable partly visible object still counts, but background
→ patterns do not. Check the named colors, shapes, identities, negations, and spatial relations, and require every
→ stated clause to hold without adding unstated conditions. Reply with exactly True or False and no other text.

Sentence: {sentence}

B.1.4 ConstructionSite

MiniCPM-V-4.5

Look at this one construction-site image and report all visible violations from the four named categories in one JSON
→ answer. Mark a category true when the image gives clear visual evidence. Do not invent details that are hidden, but do
→ not reject a plainly visible violation merely because the scene is not a close-up.

rule_1_ppe_violation: box every worker on foot who is visibly missing a hard hat or another clearly required item of
→ personal protection. A bare head is positive. Do not assume hidden footwear or clothing is missing, and do not use the
→ lack of a reflective vest in ordinary daylight by itself.

rule_2_fall_protection_violation: box every worker clearly about three metres or more above the lower level when the
→ exposed side has no effective edge protection and no safety harness is being used. Ground workers, low platforms, dirt
→ slopes, effective guardrails, and uncertain heights are negative.

rule_3_unprotected_edge_violation: box every people-accessible, continuous upper edge of a visibly deep excavation or
→ steep drop of about three metres or more when guardrails, fencing, and warning protection are absent. The image must
→ show both a substantial lower level and its hazardous upper edge; excavation activity or exposed soil alone is not
→ enough. Soil piles, excavation walls without an accessible upper edge, shallow trenches, gentle slopes, ordinary
→ uneven ground, uncertain distant cuts, and protected edges are negative. Box the hazardous upper edge and only enough
→ adjacent drop to identify it, never a worker or the whole image.

rule_4_excavator_proximity_violation: box a worker when the worker is clearly in the excavator's rear blind spot, swing
→ radius, boom or bucket path, or close enough beside the body or tracks to be reached by normal machine movement. A
→ worker merely visible in the same scene is not enough. The cab operator, a worker behind a barrier, and a worker
→ separated from the machine by a clearly safe gap are negative. Box only the endangered worker.

Use exactly this JSON shape:

```
{
  "events": {
    "rule_1_ppe_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    },
    "rule_2_fall_protection_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    },
    "rule_3_unprotected_edge_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    },
    "rule_4_excavator_proximity_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    }
  }
}
```

Change a visible category to present=true, give it a short positive reason, and add one or more worker or edge boxes. The
→ word false must always keep reason=null and boxes=[]. Coordinates are integer image-relative [x1,y1,x2,y2] values from
→ 0 to 1000 with x1 < x2 and y1 < y2. Use at most 13, 6, 3, and 2 boxes for rules 1 through 4. Return only the JSON
→ object, with no Markdown or extra text.

Qwen3.8-27B-FP8

Look at this construction-site image and report every clearly visible violation from the four categories below in one JSON
↪ answer. Most categories will usually be absent in a single image. A dangerous-looking construction scene is not enough
↪ by itself; mark a category true only when its specific target and violation are visible.

rule_1_ppe_violation means a worker on foot is visibly missing required personal protection. A clearly bare head is a
↪ violation. A worker who visibly wears a hard hat should normally be treated as compliant unless another required item
↪ is unmistakably missing for the task shown; ordinary work clothes or the absence of a reflective vest in ordinary
↪ daylight are not enough by themselves. Do not guess about small, distant, blurred, or hidden clothing, footwear, eyes,
↪ or faces. Put a tight box around each violating worker's visible body, not around the missing item.

rule_2_fall_protection_violation means a worker is clearly about three metres or more above a lower level, is exposed
↪ because effective edge protection is absent, and is not using a safety harness. Workers on the ground, on a low
↪ platform or dirt slope, behind effective guardrails, or at an uncertain height are not violations. Box each exposed
↪ worker, not the structure or edge.

rule_3_unprotected_edge_violation means an accessible upper edge borders a visibly deep excavation or steep drop of about
↪ three metres or more and lacks guardrails, fencing, or warning protection. The substantial height difference and the
↪ hazardous upper edge must both be discernible in the image rather than inferred merely because excavation work or
↪ exposed soil is present. Soil piles, excavation walls without an accessible upper edge, shallow trenches, gentle
↪ slopes, ordinary uneven ground, distant cuts with uncertain depth, and protected edges are not violations. Put one box
↪ around each continuous hazardous upper edge and only enough adjacent drop to show the hazard; do not box workers or
↪ the whole image.

rule_4_excavator_proximity_violation means a visible worker is inside the normal reach of an excavator's rear
↪ counterweight, rotating body, boom, or bucket. A still photograph need not show actual motion: count a worker who is
↪ clearly within the machine's ordinary swing or travel zone, including directly beside its rear or tracks in the active
↪ work area. Do not count the cab operator, a worker behind a barrier, a worker separated by a clearly safe gap, or
↪ simple distant co-appearance. Box each endangered worker, not the excavator or a combined worker-machine area.

Return exactly one JSON object whose "events" object contains "rule_1_ppe_violation", "rule_2_fall_protection_violation",
↪ "rule_3_unprotected_edge_violation", and "rule_4_excavator_proximity_violation" exactly once. A visible event is
↪ {"present":true,"reason":"short visible evidence","boxes":[[x1,y1,x2,y2]]}; an absent or uncertain event is
↪ {"present":false,"reason":null,"boxes":[]}. Coordinates are integer image-relative values from 0 to 1000 with $x_1 < x_2$
↪ and $y_1 < y_2$. Return at most 13, 6, 3, and 2 boxes for rules 1 through 4. Return no Markdown or text outside the JSON
↪ object.

B.2 Earlier Prompt Comparisons

Figures below reproduce earlier prompt comparisons on training subsets: 300 expressions for gRefCOCO and 400 examples each for CUB and NLVR2. Construction uses its recorded training selection. The source label Construction10K refers to ConstructionSite. These curves use the earlier task metrics (including gRefCOCO F1), so their absolute values are not directly comparable with the final routing table. They document prompt choice, not router ablations or new test-time tuning. The experiment ledger retains the input subset and metric for every point.

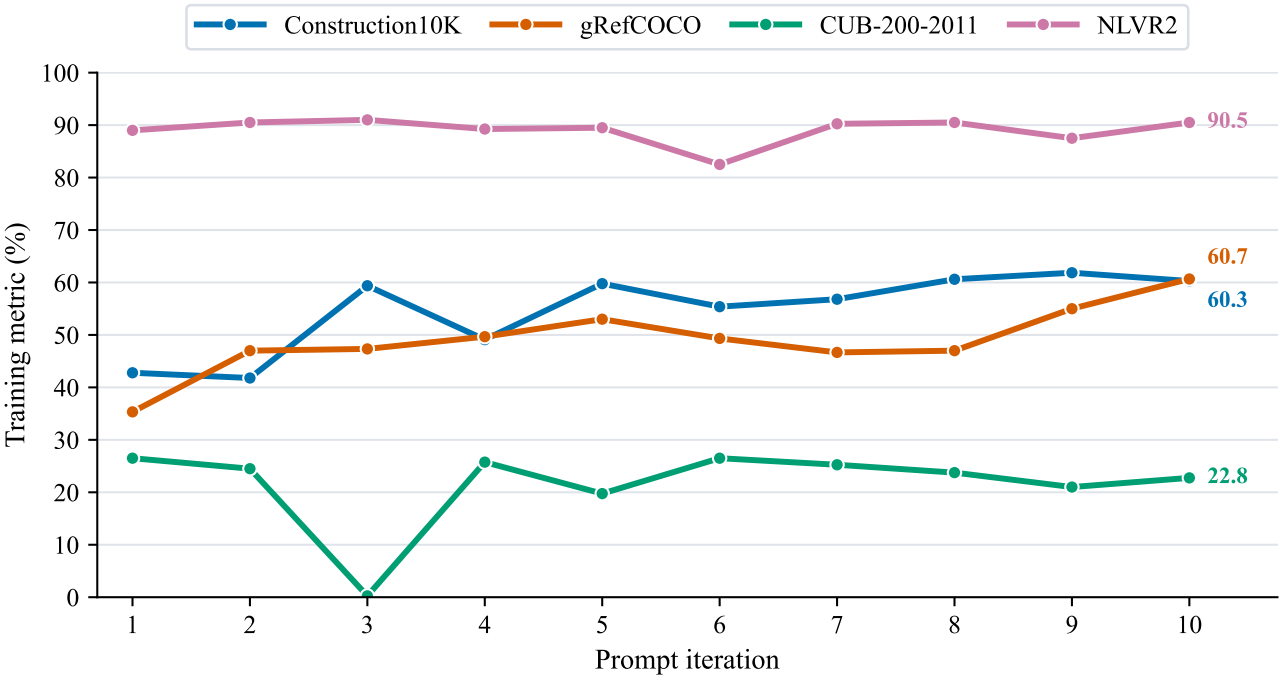


Figure S2: Earlier MiniCPM prompt comparisons from the recorded experiment ledger.

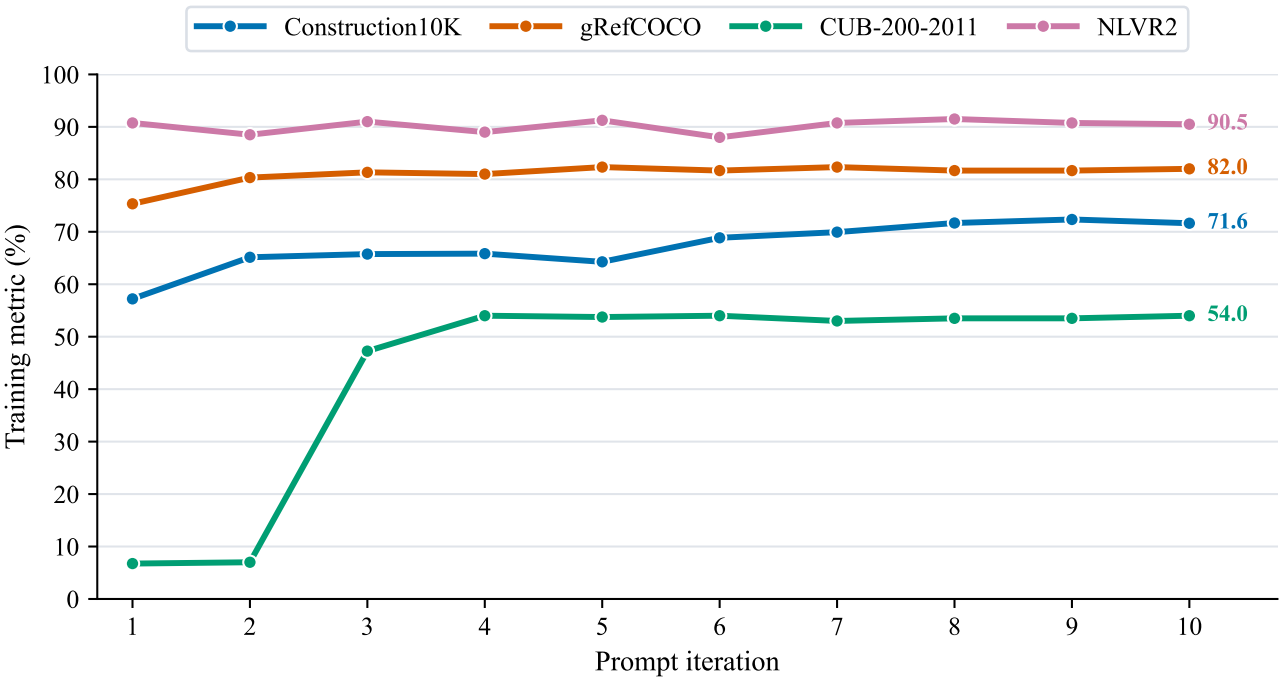


Figure S3: Earlier Qwen prompt comparisons from the same ledger.

B.3 Feature Construction and Supervision

Image features contain two ordered 768-dimensional slots and two availability indicators. Single-image tasks use a fixed gray placeholder in the second slot with its indicator off. Image-only tasks omit the task-text branch. Specialist hidden-state and confidence features are standardized with training statistics; specialist-specific pooling follows the archived feature structure. Each active branch uses Linear+LayerNorm to width 128. Concatenation feeds an MLP with hidden width 128 and GELU, followed by four logits ordered as both correct, rescue, harm and both wrong.

The target for input i is the empirical state distribution across its m_i correctness units:

$$q_i = \frac{1}{m_i} \sum_{j=1}^{m_i} [c_{S,ij}c_{L,ij}, (1 - c_{S,ij})c_{L,ij}, c_{S,ij}(1 - c_{L,ij}), (1 - c_{S,ij})(1 - c_{L,ij})]^\top. \quad (S3)$$

Here $m_i = 1$ for CUB, NLVR2 and whole-expression gRefCOCO correctness, and $m_i = 4$ for construction events. Thus a gRefCOCO expression referring to several objects still receives one state, not a box-averaged soft label. We minimize cross-entropy with q_i and rank by the rescue-minus-harm logit contrast. With softmax probabilities p , this contrast equals $\log(p_R/p_H)$, rather than $p_R - p_H$. The four-state objective does not make the evaluated score an estimate of expected gain.

C Operating-Point Selection and Complete Results

Two stages use validation labels. First, training checkpoints and the common configuration maximize a ranking-area criterion. With $b_k = 0.05k$, define

$$\text{Score}_{25} = \frac{1}{0.25} \sum_{k=1}^5 \frac{M(b_{k-1}) + M(b_k)}{2} (b_k - b_{k-1}). \quad (S4)$$

At each b_k , exactly $\lfloor b_k n_{\text{val}} \rfloor$ validation examples with the highest scores are delegated. Stable sample order breaks ties. The selection score averages equally over four pairs per task, then equally over four tasks. This is a validation selection score, not a pooled test accuracy across incomparable metrics.

Second, the selected router is assigned thresholds for validation call budgets $b \in \{0.01, \dots, 0.99\}$. A threshold uses $s \geq t_b$ when all boundary ties fit the budget, and $s > t_b$ otherwise. The candidate with the highest validation task metric wins; equal metrics favor fewer actual calls and then a lower nominal budget. Confidence uses $s = -\max p$ or the negative maximum detection confidence. Missing confidence has priority over valid scores. Both methods follow the same threshold procedure.

The chosen threshold value and comparison operator are fixed on test inputs. A budget of 25% therefore means a threshold chosen for a validation call budget, not a guarantee that exactly 25% of test inputs will call the VLM. The saved final search excludes 0% and 100%; standalone scores remain separate endpoints. These exclusions explain why an interior selected policy can underperform the specialist. No maximum is chosen from test curves.

Table S4 reports all validation and corresponding test scores, with actual calls for each split. Table S3 records the budgets and numerical cutoffs behind them. Score and call differences are percentage points computed from unrounded fractions. For example, changing calls from 92.21% to 48.77% gives a 43.44-point decrease, not a relative percent reduction. Green marks improvement and red degradation: higher task scores and fewer calls are better.

Table S3: Selected nominal validation budgets and score cutoffs. The operator is part of each stored threshold. Displayed cutoffs are rounded; the archived JSON retains full precision. Budget is a percentage, not the numerical threshold.

Task	Specialist	VLM	Conf. budget	Conf. cutoff	Learned budget	Learned cutoff
CUB-200-2011	ResNet	M	1	≥ -0.305811	1	≥ -0.590226
	ResNet	Q	1	≥ -0.305811	1	≥ 0.600258
	GLSim	M	1	≥ -0.48844	3	≥ -1.56001
	GLSim	Q	1	≥ -0.48844	1	≥ 4.02929
gRefCOCO	GroundingDINO	M	99	≥ -0.916231	34	≥ 6.07611
	GroundingDINO	Q	99	≥ -0.916231	42	≥ 5.46601
	InstanceVG	M	65	≥ -0.987771	23	≥ -5.38178
	InstanceVG	Q	71	≥ -0.994979	56	≥ -5.43594
NLVR2	BEiT3	M	86	> -0.999739	85	≥ -5.67533
	BEiT3	Q	86	> -0.999739	77	≥ -3.18357
	ViLT	M	96	≥ -0.99984	99	≥ -4.51107
	ViLT	Q	96	≥ -0.99984	95	≥ -7.2186
ConstructionSite	YOLO26x	M	94	≥ -0.490234	49	≥ -2.5235
	YOLO26x	Q	93	≥ -0.453369	50	≥ -1.19634
	RT-DETR-X	M	98	≥ -0.422105	78	≥ -2.93597
	RT-DETR-X	Q	99	≥ -0.441623	83	≥ -0.421618

Table S4: Complete four-way comparison. Scores and actual call rates are percentages. Thresholds selected on the 1–99% validation grid transfer unchanged to test. Parentheses give learned-minus-confidence differences in percentage points before rounding. Green marks improvement and red degradation; higher scores and fewer calls are better. Bold marks the higher routed score; † exceeds both test endpoints. M: MiniCPM-V-4.5; Q: Qwen3.8-27B-FP8.

Task	Specialist	VLM	Validation						Test					
			spec.	vlm.	conf.	call _{conf}	learned†	call _{learned} ↓	spec.	vlm.	conf.	call _{conf}	learned†	call _{learned} ↓
CUB-200-2011	ResNet	M	98.58	27.75	98.50	1.00	98.58 (↑ 0.08)	1.00 (=0.00)	81.43	26.56	79.93	11.36	81.38 (↑ 1.45)	2.88 (↓ 8.47)
	ResNet	Q	98.58	55.08	98.25	1.00	98.58 (↑ 0.33)	1.00 (=0.00)	81.43	54.82	81.58	11.36	81.34 (↓ 0.24)	1.43 (↓ 9.92)
	GLSim	M	98.92	27.75	98.50	1.00	99.00 (↑ 0.50)	3.00 (↑ 2.00)	90.16	26.56	87.69	8.70	89.80 (↑ 2.11)	4.49 (↓ 4.21)
	GLSim	Q	98.92	55.08	98.58	1.00	98.83 (↑ 0.25)	1.00 (=0.00)	90.16	54.82	88.78	8.70	90.06 (↑ 1.28)	1.66 (↓ 7.04)
gRefCOCO	GroundingDINO	M	56.05	64.12	63.72	99.00	68.01 (↑ 4.29)	33.99 (↓ 65.00)	26.69	50.01	49.68	98.74	45.62 (↓ 4.06)	43.52 (↓ 55.22)
	GroundingDINO	Q	56.05	80.60	79.94	99.00	81.53 (↑ 1.59)	42.00 (↓ 57.00)	26.69	74.47	73.87	98.74	66.10 (↓ 7.77)	57.46 (↓ 41.28)
	InstanceVG	M	75.03	64.12	77.17	64.99	77.30 (↑ 0.13)	23.00 (↓ 42.00)	66.95	50.01	61.88	52.09	66.72 (↑ 4.84)	30.90 (↓ 21.19)
	InstanceVG	Q	75.03	80.60	82.02	71.00	80.81 (↓ 1.20)	56.00 (↓ 15.00)	66.95	74.47	72.43	68.39	75.28 † (↑ 2.85)	53.56 (↓ 14.83)
NLVR2	BEiT3	M	79.82	89.90	90.05	85.88	89.95 (↓ 0.10)	84.99 (↓ 0.89)	80.00	90.74	90.77	85.80	90.64 (↓ 0.13)	84.66 (↓ 1.14)
	BEiT3	Q	79.82	90.50	90.58	85.88	90.95 (↑ 0.37)	77.00 (↓ 8.88)	80.00	90.06	90.24	85.80	90.88 † (↑ 0.64)	76.66 (↓ 9.14)
	ViLT	M	75.57	89.90	89.97	95.99	89.82 (↓ 0.16)	99.00 (↑ 3.01)	76.42	90.74	90.64	95.93	90.71 (↑ 0.06)	98.97 (↑ 3.04)
	ViLT	Q	75.57	90.50	90.59	95.99	90.42 (↓ 0.17)	94.99 (↓ 1.00)	76.42	90.06	89.96	95.93	90.00 (↑ 0.04)	94.86 (↓ 1.08)
ConstructionSite	YOLO26x	M	18.25	20.67	22.64	93.87	29.01 (↑ 6.37)	48.93 (↓ 44.94)	18.13	22.36	25.23	92.54	25.82 † (↑ 0.59)	47.20 (↓ 45.34)
	YOLO26x	Q	18.25	25.45	26.43	92.87	33.13 (↑ 6.70)	49.93 (↓ 42.94)	18.13	27.00	25.97	92.21	29.88 † (↑ 3.90)	48.77 (↓ 43.44)
	RT-DETR-X	M	7.64	20.67	20.57	97.86	21.74 (↑ 1.17)	77.89 (↓ 19.97)	7.89	22.36	21.61	97.90	23.00 † (↑ 1.39)	78.83 (↓ 19.07)
	RT-DETR-X	Q	7.64	25.45	24.86	98.86	27.20 (↑ 2.34)	82.88 (↓ 15.98)	7.89	27.00	26.82	98.60	26.69 (↓ 0.13)	83.12 (↓ 15.48)

D Complete Rescue–Harm Accounting

Table S5: Test outcome counts at the learned operating point. BC/BW: both correct/wrong; R/H: rescue/harm. Selected rescues, selected harms, missed rescues and avoided harms are counted separately. Construction counts 12,016 events, with routing over 3,004 images. Metric gains are recomputed; macro-F1 is not an event-count ratio.

Task	Specialist	VLM	BC	R	H	BW	Sel. R	Sel. H	Miss R	Avoid H	Gain (pp)
CUB-200-2011	ResNet	M	1,307	232	3,411	844	11	14	221	3,397	-0.052
	ResNet	Q	2,811	365	1,907	711	5	10	360	1,897	-0.086
	GLSim	M	1,433	106	3,791	464	9	30	97	3,761	-0.362
	GLSim	Q	3,032	144	2,192	426	0	6	144	2,186	-0.104
gRefCOCO	GroundingDINO	M	6,911	10,723	2,500	15,129	7,140	463	3,583	2,037	+18.935
	GroundingDINO	Q	7,441	18,820	1,970	7,032	14,636	738	4,184	1,232	+39.412
	InstanceVG	M	15,007	2,627	8,602	9,027	1,570	1,652	1,057	6,950	-0.233
	InstanceVG	Q	20,838	5,423	2,771	6,231	4,361	1,423	1,062	1,348	+8.332
NLVR2	BEiT3	M	10,421	2,225	729	562	2,140	657	85	72	+10.641
	BEiT3	Q	10,305	2,247	845	540	2,060	544	187	301	+10.878
	ViLT	M	9,940	2,706	711	580	2,695	704	11	7	+14.286
	ViLT	Q	9,761	2,791	890	495	2,767	875	24	15	+13.575
ConstructionSite	YOLO26x	M	8,176	1,198	2,109	533	1,158	850	40	1,259	+7.692
	YOLO26x	Q	9,508	1,282	777	449	1,249	337	33	440	+11.746
	RT-DETR-X	M	6,060	3,314	1,945	697	3,304	1,470	10	475	+15.109
	RT-DETR-X	Q	7,329	3,461	676	550	3,448	576	13	100	+18.806

For accuracy tasks, selected rescues minus selected harms gives the change in correct answers. Both-correct and both-wrong calls use budget without changing correctness. The four available pools describe model complementarity; their sizes are not a router’s success rate.

InstanceVG/Q selects 4,361 rescues and 1,423 harms. Relative to the specialist, this is 2,938 additional correct answers among 35,263 expressions, or 8.332 percentage points. Relative to always using Q, the router avoids 1,348 harmful replacements while missing 1,062 rescues; the 286-example difference gives a 0.811-point gain. In contrast, InstanceVG/M selects 1,570 rescues and 1,652 harms, so recoverability exists without a net gain over the specialist at the selected point.

Construction has four correctness units per image but only one delegation decision. YOLO26x/Q selects 1,249 rescuable events and incurs 337 harmful changes; the recomputed macro-F1 gain is 11.746 points. Dividing the event net gain by the image count would be incorrect, as would treating event correctness as an oracle bound for macro-F1.

E Budget Curves and Matched-Call Diagnostics

Table S6: Low-budget test comparison in 100Score₂₅. Transferred thresholds use validation-selected budgets; exact-budget ranking selects the same number of test inputs for each router at each budget without reading test labels. The latter is a batch-ranking diagnostic, not a new deployed operating point. All thresholds, settings and ranking scores remain fixed.

Task	Specialist	VLM	Transferred thresholds			Exact-budget test ranking		
			Conf.	Learned	Difference	Conf.	Learned	Difference
CUB	ResNet	M	64.729	77.632	+12.903	78.959	79.676	+0.716
	ResNet	Q	73.162	80.532	+7.370	81.015	80.637	-0.378
	GLSim	M	72.534	87.402	+14.869	85.198	88.091	+2.893
	GLSim	Q	78.605	89.185	+10.580	86.912	89.356	+2.444
gRefCOCO	GroundingDINO	M	27.538	33.085	+5.547	27.866	33.525	+5.659
	GroundingDINO	Q	28.625	35.497	+6.872	29.324	37.352	+8.028
	InstanceVG	M	66.393	67.592	+1.199	65.900	67.773	+1.873
	InstanceVG	Q	66.407	72.810	+6.402	66.114	71.368	+5.254
NLVR2	BEiT3	M	83.390	84.256	+0.866	83.588	84.142	+0.554
	BEiT3	Q	83.440	84.363	+0.923	83.613	84.253	+0.640
	ViLT	M	79.874	80.051	+0.177	80.052	80.152	+0.100
	ViLT	Q	79.887	80.259	+0.372	80.062	80.375	+0.313
Construction	YOLO26x	M	17.698	21.733	+4.036	17.166	21.580	+4.414
	YOLO26x	Q	18.055	21.316	+3.261	17.933	21.300	+3.367
	RT-DETR-X	M	8.325	9.727	+1.401	8.357	9.796	+1.438
	RT-DETR-X	Q	8.911	10.227	+1.316	8.908	10.460	+1.552

The transferred-threshold columns use validation-selected operating rules. The exact-budget columns instead rank each test batch by saved scores and choose exactly $\lfloor bn_{\text{test}} \rfloor$ calls at each of the six Score₂₅ budgets; test labels never determine ordering. This separates ranking quality from threshold transfer, but assumes a batch is available for sorting. It does not replace the final operating points in Table S4.

Learned routing improves transferred-threshold test area on all sixteen pairs and exact-budget area on fifteen. ResNet/Q is the latter exception (80.637 versus 81.015). For GroundingDINO, exact-budget improvements are 5.659–8.028 points despite the poorer transferred maximum-performance operating points. All curves below contain the same 99 saved validation thresholds plus the specialist and VLM endpoints, without smoothing; dots mark validation-selected points.

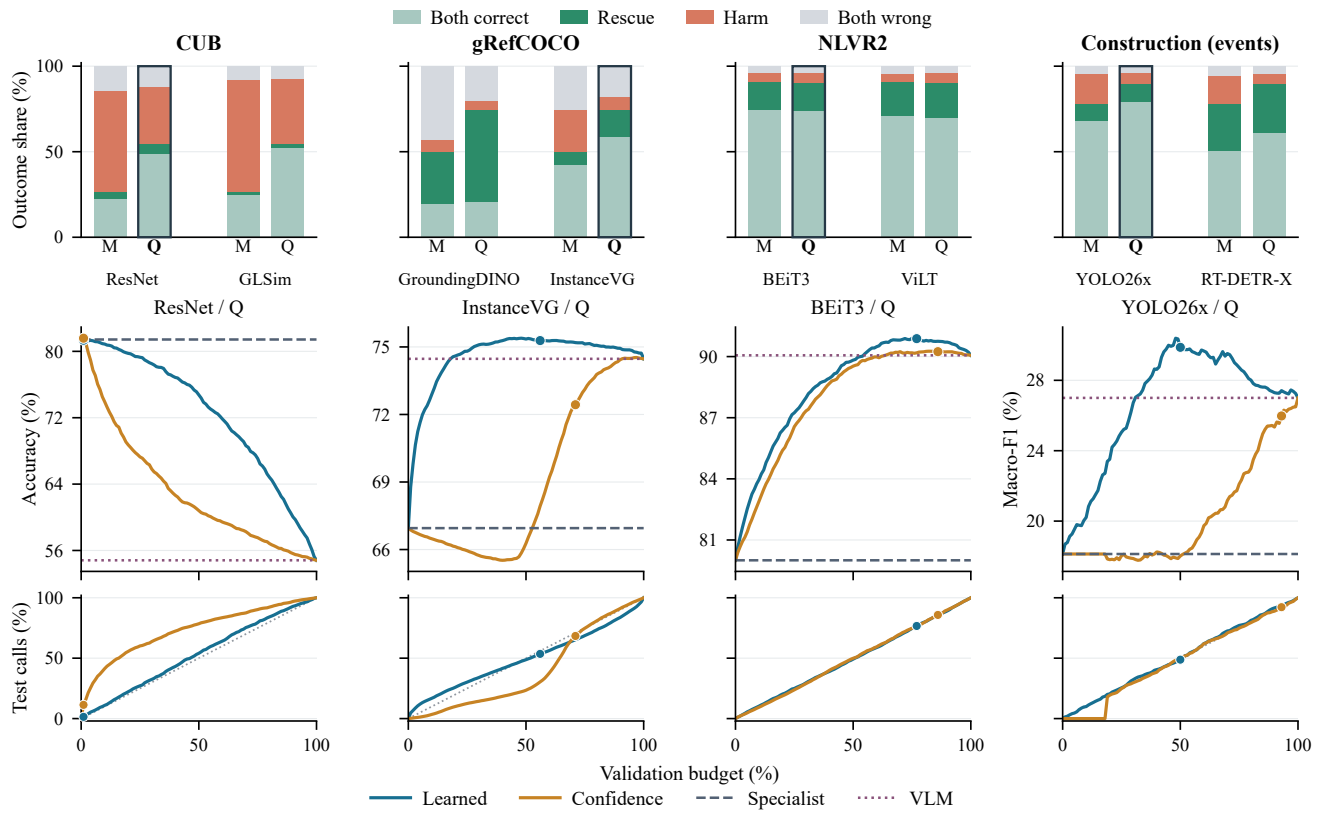


Figure S4: Complete three-row view: test outcome shares, test performance and actual test calls. The horizontal axis on the bottom two rows is nominal validation budget. Outlined bars identify the four example pairs below. Construction outcomes count events, while calls count images.

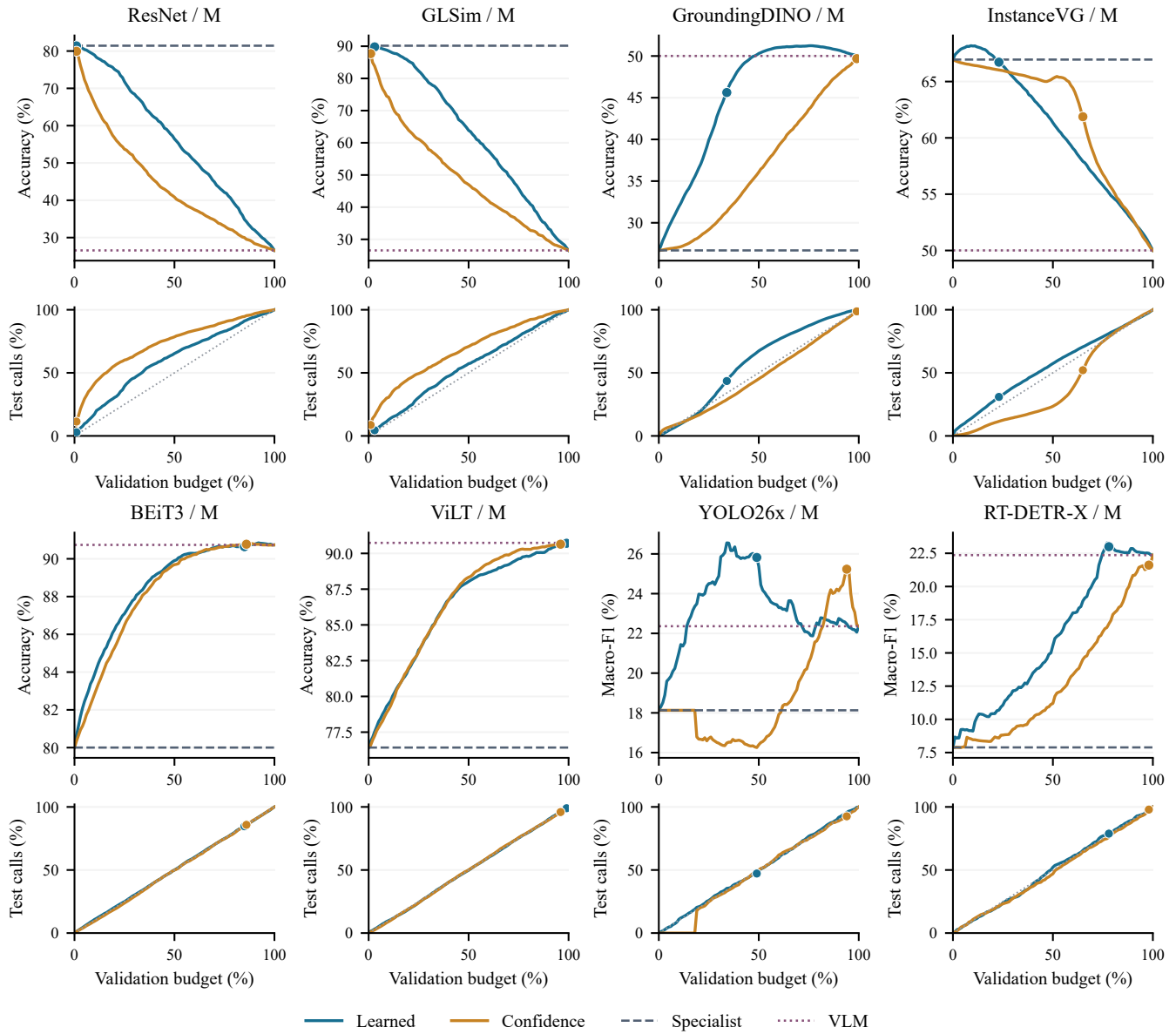


Figure S5: All eight specialist/MiniCPM pairs. Upper panels show test performance; lower panels show actual test calls under the same nominal validation budgets. Colored points use the validation-selected thresholds.

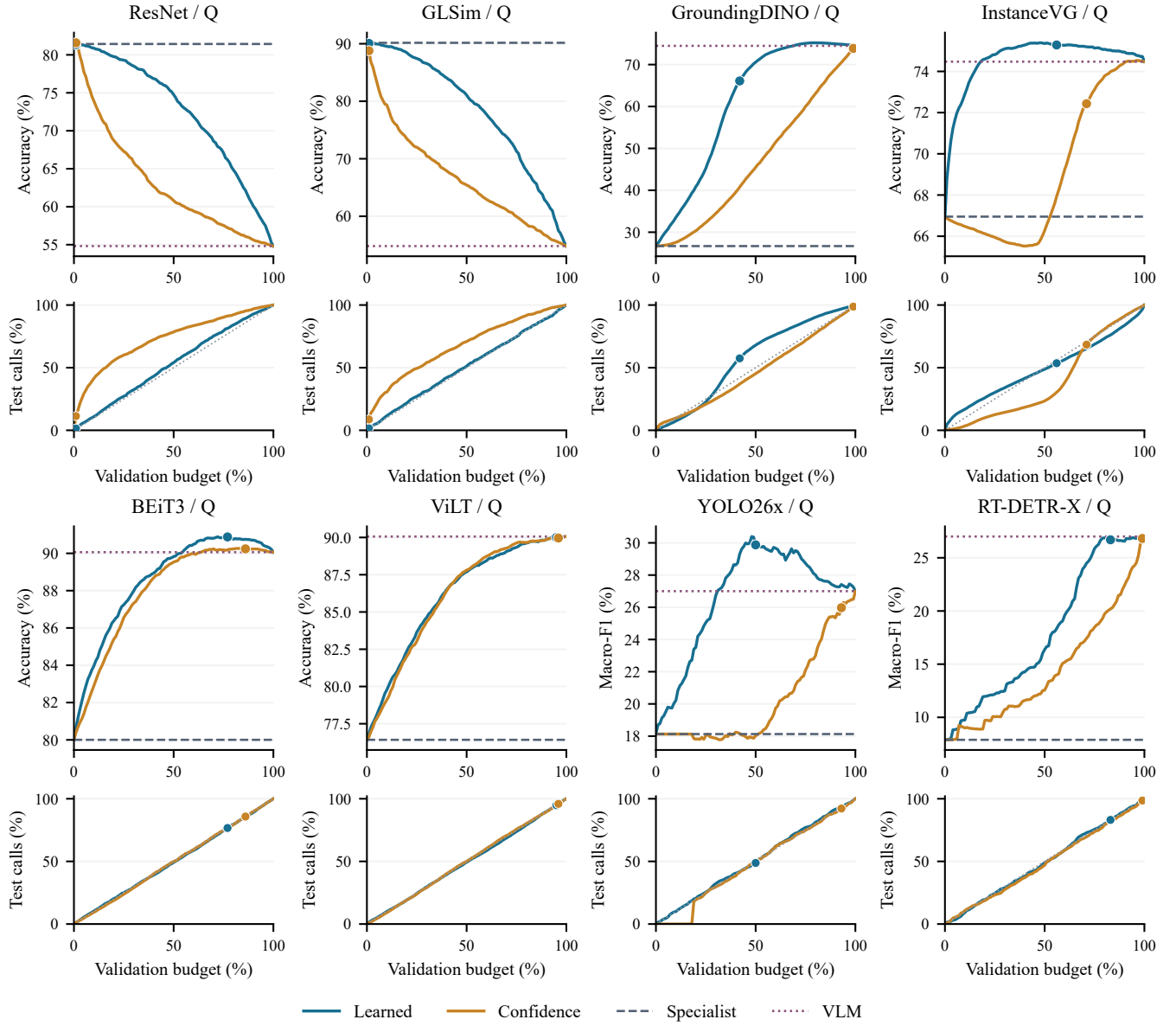


Figure S6: All eight specialist/Qwen pairs, with the same conventions. The identity line in call panels marks equality of validation budget and test call rate, not an enforced test constraint.

F Shared-Recipe Sensitivity and Earlier Ablations

Table S7: Validation sensitivity in the common-recipe search. The selection score is the task-balanced mean of validation 100Score₂₅, not a pooled accuracy. Joint: four-state cross-entropy; Gain: gain regression; Error: specialist-error prediction. These eighteen settings all contributed to selection, so the differences are comparisons used to choose the final settings. Bold marks the selected recipe.

Setting	Student	Width	Learning rate	Depth	Target	Sampling	Selection score
1	4M	512	0.001	90%	Joint	Natural	66.2818
2	4M	512	0.001	30%	Joint	Natural	66.0601
3	4M	512	0.001	60%	Joint	Natural	66.1705
4	1M	512	0.001	90%	Joint	Natural	66.3764
5	2M	512	0.001	90%	Joint	Natural	66.1689
6	8M	512	0.001	90%	Joint	Natural	66.5048
7	8M	32	0.001	90%	Joint	Natural	66.5674
8	8M	64	0.001	90%	Joint	Natural	66.7630
9	8M	128	0.001	90%	Joint	Natural	66.7768
10	8M	256	0.001	90%	Joint	Natural	66.6584
11	8M	128	0.0001	90%	Joint	Natural	66.6107
12	8M	128	0.0003	90%	Joint	Natural	66.6167
13	8M	128	0.003	90%	Joint	Natural	66.7871
14	8M	128	0.01	90%	Joint	Natural	66.3415
15	8M	128	0.003	90%	Gain	Natural	66.5283
16	8M	128	0.003	90%	Error	Natural	65.2940
17	8M	128	0.003	90%	Joint	Weighted	66.4918
18	8M	128	0.003	90%	Joint	Balanced	66.3128

The search is sequential. Valid one-factor groups are settings 1–3 (depth); 1 and 4–6 (size); 6–10 (width); 9 and 11–14 (learning rate); 13 and 15–16 (supervision); and 13 and 17–18 (sampling). The 0.003 versus 0.001 learning-rate difference is only 0.0103 validation selection-score points.

Four-state cross-entropy, direct gain regression and specialist-error prediction share the final configuration in their comparison. The weighted-loss and balanced-sampling comparisons also use that configuration. This supports comparing the tested objectives within the choice of training settings; it does not establish that each feature branch is necessary. A matched removal of image, text or hidden features from the exact final recipe remains absent.

Table S8: Comparisons around earlier pair-specific configurations, all paired with Q. These are not removals from the new shared recipe. V-area is 100Score₂₅ on validation. Test threshold performance and calls use the validation 25% threshold; Test rank fixes test calls to $\lfloor 0.25n \rfloor$. Values are percentages. Each block retains its own objective and training configuration. I/T/H/C/P denote image, text, hidden state, confidence and prediction summary.

Pair	Earlier setting / change	V-area	Test threshold	Test calls	Test rank 25%
ResNet/Q	P+C reference	92.800	68.295	60.49	80.911
	Remove C	97.908	80.652	24.77	80.652
	Remove P	92.108	69.503	54.73	78.944
GroundingDINO/Q	I+T+H+C reference	66.509	43.034	21.27	45.441
	Remove I	66.675	43.431	21.80	45.419
	Remove T	66.438	44.171	22.87	45.532
	Remove H	66.213	44.636	23.80	45.373
	Remove C	66.462	42.835	21.18	45.385
BEiT3/Q	P+C reference	83.834	87.092	25.40	86.956
	Remove C	81.813	80.003	0.00	83.555
	Remove P	83.138	86.568	23.87	86.719
YOLO26x/Q	P reference, width 512	24.930	25.038	18.34	26.328
	Width 32	24.807	29.177	43.21	26.310

F.1 Hidden-State Depth

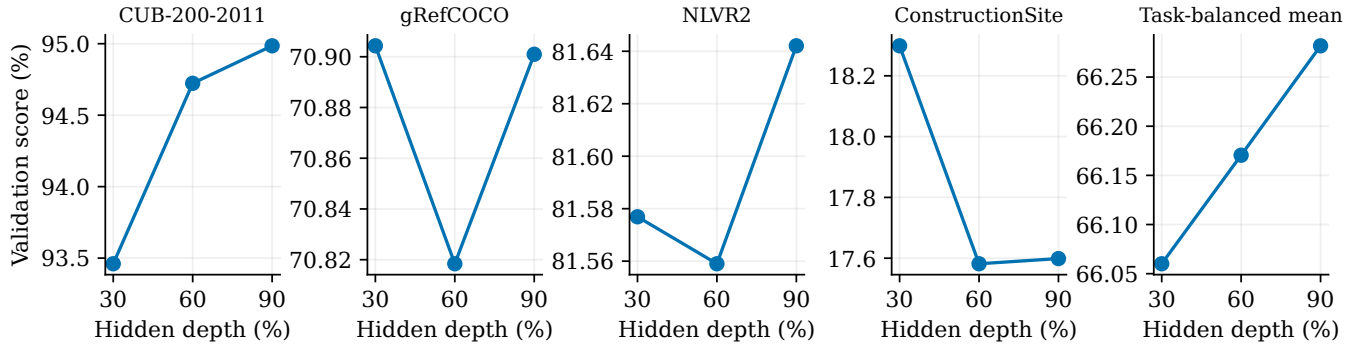


Figure S7: Validation low-budget scores at 30%, 60% and 90% relative hidden depth. Each task averages its four model pairs, and the final panel averages the four tasks equally. All three settings use the 4M student, width 512, learning rate .001, four-state targets and natural sampling. Thus, only hidden depth changes. These are the first stage of the sequential search, not additional sweeps around the final 8M setting.

Relative depth makes models with different layer counts comparable: the chosen position is rounded upward within the model’s recorded sequence of hidden states. Larger depth need not improve every task, so the shared depth is chosen by the task-balanced validation mean. The curve is shown without smoothing or uncertainty bands because each setting uses seed 42.

F.2 What the Five Input Ablations Establish

The five input types are image features (I), task text (T), specialist hidden state (H), confidence (C) and prediction summary (P). Table S8 contains earlier one-input removals: I/T/H/C on GroundingDINO and C/P on ResNet and BEiT3. These comparisons cover all five input types across their applicable model pairs, but do not form a five-way removal study of the final shared router. Image-only tasks have no task-text branch. Removing an input also changes the head’s input dimension.

The main shared-setting study deliberately retains image and hidden features and includes text whenever the task has it. Consequently, the existing results do not establish that each of its five inputs is individually necessary. The code supplies separate final-setting removal configurations for future runs; unrun settings have no reported scores. Table S8 should be interpreted only against the reference within each block. Earlier experiments retain their original references. GroundingDINO’s apparent threshold-score gain comes with more calls and lower matched-call performance. These are not final-recipe removals.

G Additional Data-Volume and Joint-Training Results

G.1 Router-Training Data Volume

Table S9: Earlier training-data comparisons. Each pair retains its exploratory reference configuration; only router-training fraction changes. Within each fraction: test metric, actual test calls, and test metric at exactly 25% ranked calls. The first two use a validation-selected 25% threshold. These are not final shared-recipe ablations. All entries are percentages.

Task	Specialist	VLM	25% training			50% training			100% training		
			Metric	Calls	Rank	Metric	Calls	Rank	Metric	Calls	Rank
CUB-200-2011	ResNet	M	78.49	25.87	78.84	79.05	25.49	79.31	78.72	25.58	79.13
	ResNet	Q	68.74	60.39	80.60	69.43	59.98	80.96	68.29	60.49	80.91
	GLSim	M	86.74	24.82	86.74	87.00	24.99	87.00	86.59	25.51	86.80
	GLSim	Q	87.97	23.66	87.76	87.64	24.53	87.63	86.85	25.08	86.85
gRefCOCO	GroundingDINO	M	38.16	24.84	38.20	36.59	22.03	37.58	36.93	22.85	37.58
	GroundingDINO	Q	45.63	24.86	45.71	43.46	22.01	45.37	43.03	21.27	45.44
	InstanceVG	M	65.23	34.79	66.53	65.67	31.24	66.28	65.48	33.95	66.56
	InstanceVG	Q	70.98	10.64	70.20	74.17	48.24	72.08	70.90	12.33	70.22
NLVR2	BEiT3	M	86.38	25.39	86.30	86.83	24.39	86.90	87.08	26.12	86.89
	BEiT3	Q	87.18	25.30	87.14	87.21	25.41	87.03	87.09	25.40	86.96
	ViLT	M	83.24	24.07	83.43	83.38	24.70	83.45	83.27	24.21	83.47
	ViLT	Q	83.36	24.43	83.47	83.09	23.61	83.41	83.30	24.09	83.48
ConstructionSite	YOLO26x	M	22.26	18.34	22.25	22.26	18.34	22.25	22.28	18.87	22.27
	YOLO26x	Q	25.04	18.34	26.33	25.04	18.34	26.33	25.04	18.34	26.33
	RT-DETR-X	M	12.16	26.90	12.10	11.14	25.80	11.16	11.41	23.34	11.42
	RT-DETR-X	Q	14.47	28.93	14.35	15.00	26.80	14.99	14.98	25.40	15.03

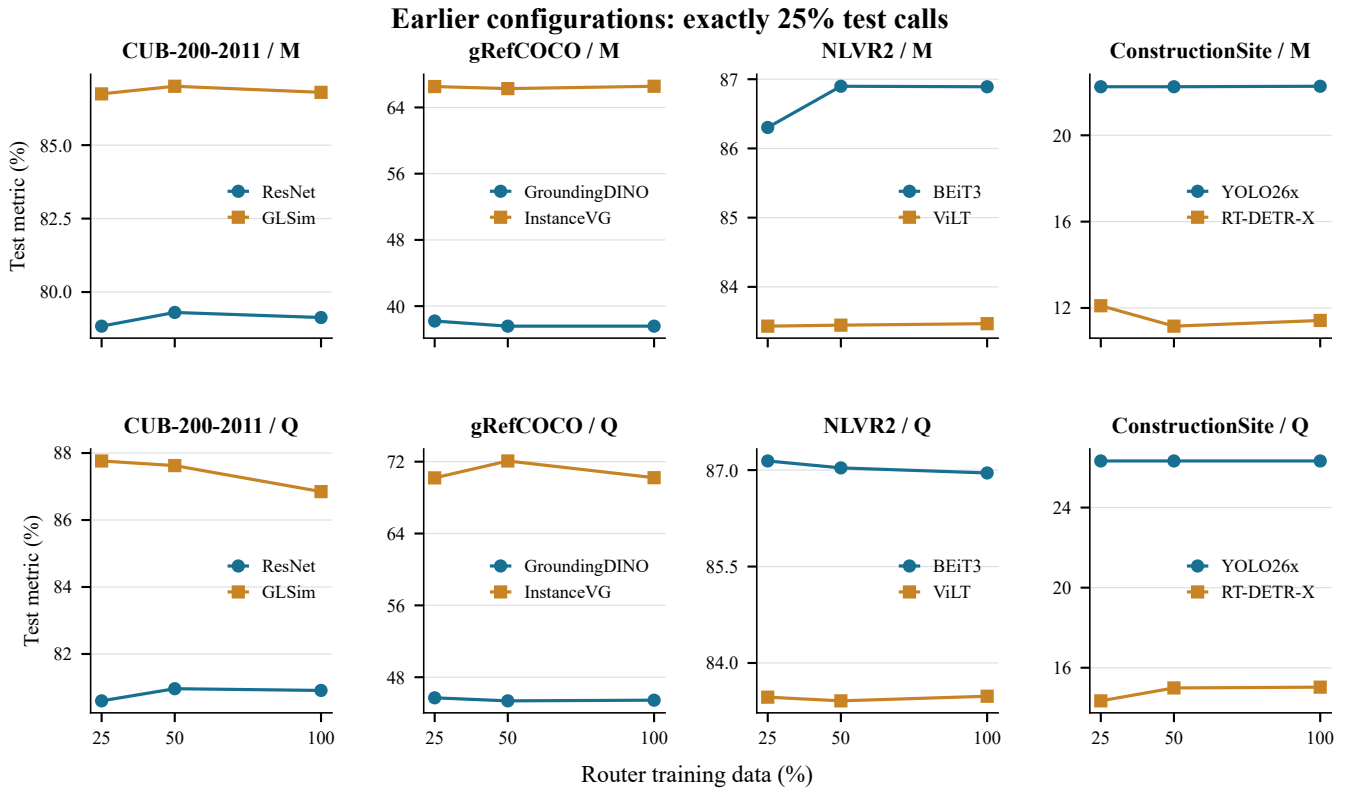


Figure S8: Data-volume comparison at exactly 25% test calls, using archived ranking results. Each pair’s reference configuration is unchanged across its three training fractions. This view removes the varying test call volumes in the transferred-threshold comparison. Metrics are accuracy except construction macro-F1.

Increasing labeled router data is different from retraining a specialist with more images. Each curve changes only the router’s training fraction around an earlier pair-specific reference. Results need not be monotonic. Neither these curves nor the transferred-threshold scores establish a scaling law for the final shared recipe.

G.2 Single-Task and Joint Training

Table S10: Earlier single-task versus joint-training comparison. Both use 4M students, image/text inputs and specialist-error supervision. Metrics use validation-selected 25% thresholds; actual call rates are not present in this comparison file. Differences are joint minus single in percentage points. All tasks were seen during training.

Task	Specialist	VLM	Single	Joint	Difference (pp)
CUB-200-2011	ResNet	M	71.194	70.797	-0.397
	ResNet	Q	76.200	77.822	+1.622
	GLSim	M	75.923	77.459	+1.536
	GLSim	Q	83.017	84.484	+1.467
gRefCOCO	GroundingDINO	M	35.981	37.005	+1.024
	GroundingDINO	Q	42.240	45.992	+3.752
	InstanceVG	M	63.942	62.437	-1.506
	InstanceVG	Q	72.249	70.530	-1.719
NLVR2	BEiT3	M	83.468	83.095	-0.373
	BEiT3	Q	83.727	83.375	-0.352
	ViLT	M	80.469	80.261	-0.208
	ViLT	Q	80.455	80.677	+0.222
ConstructionSite	YOLO26x	M	18.033	18.362	+0.329
	YOLO26x	Q	19.009	19.603	+0.594
	RT-DETR-X	M	10.367	11.260	+0.893
	RT-DETR-X	Q	10.483	12.038	+1.555

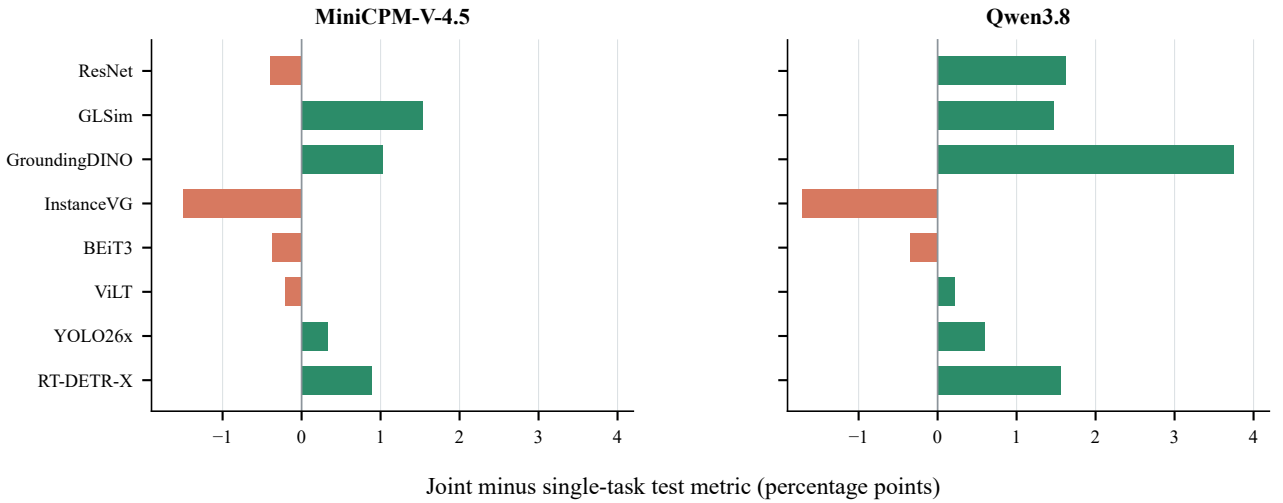


Figure S9: Change from independently trained routers to joint training under the earlier training setup. Scores use thresholds chosen for 25% validation calls. Positive changes are green and negative changes orange. This experiment includes all evaluated tasks during training.

Sharing training does not uniformly improve every pair. The figure and table compare their recorded points selected for 25% validation calls; since the summary does not provide actual call rates, the differences cannot be attributed solely to ranking quality at matched calls. These results concern sharing across already-seen tasks and are not evidence for zero-shot routing on an unseen task.

H Case Study

(a) Rescue



“all individuals”

Targets: 4 | Call VLM

Model	Matched	Boxes	Correct
InstanceVG	2	5	No
Qwen3.8	4	4	Yes

(b) Harm



“white man”

Targets: 0 | Call VLM

Model	Matched	Boxes	Correct
InstanceVG	0	0	Yes
Qwen3.8	0	3	No

(c) Missed rescue



“man with curly hair and black shirt, back row on left”

Targets: 0 | Keep specialist

Model	Matched	Boxes	Correct
InstanceVG	0	1	No
Qwen3.8	0	0	Yes

Figure S10: Three test expressions for the selected InstanceVG/Q policy: rescue, harmful replacement and missed rescue. Within each outcome group, the lowest sample identifier was chosen before inspecting the image. Photos are unmodified; unavailable predicted boxes are not reconstructed. Panels (a) and (c) refer to different expressions on the same image, not independent images.

A rescue illustrates a selected VLM prediction satisfying the strict localization rule when the specialist fails. A harmful replacement illustrates the reverse. A missed rescue shows that complementary information can remain unused. The examples illustrate error types; the full counts in Sec. D, rather than these three panels, establish frequency. The method figure in the main paper uses a separate saved gRefCOCO validation example and its original query; it is an architectural illustration rather than an additional test case.

I Computational-Cost Records and Their Limits

The study in main-paper Sec. 4.6 retains the main-table checkpoints, seed 42 and validation thresholds. Selected test routing decisions match the archived main-table decisions exactly. Separate A800 timing probes use 64 inputs per component at batch one, with two VLM warm-up inputs excluded. Learned-routing costs include the distilled encoders and routing head. Confidence costs equal specialist cost plus actual test call rate times mean VLM cost; learned routing additionally includes encoder and head costs. These are serial estimates from component measurements, not directly measured full-system latency. FLOPs count a multiply-add as two operations, using four specialist inputs and eight encoder/head and VLM inputs; VLM cost includes visual encoding and prefill plus mean output length times single-token decoding cost. Unsupported operators are omitted. Both policies share the same mean VLM cost, so selected-input length differences are not measured. Specialist hidden features are cached and their extra extraction cost is not separately measured; confidence-threshold overhead is also not separately timed. Model loading and warm-up are excluded. The 404 curve points and four selected points are supplied in `data/main_table_cost_qwen.20260912/`.

I.1 Earlier Partial-Cost Records

Average deployment computation should include all three components,

$$\bar{C} = \frac{1}{n} \sum_i [C_S(x_i) + C_R(x_i) + a_i C_L(x_i)]. \quad (\text{S5})$$

Frozen router encoders still run at inference. The archived counts cover seven specialists over 32 inputs each, plus one fixed CUB prefill probe for each VLM. InstanceVG, full decoding, some operators and aligned per-input router costs are missing. No measured system-wide savings follow from these partial counts.

Table S11: Archived partial operation counts in GFLOPs. Specialist means use 32 inputs; the range is across those inputs. Each VLM row is one fixed CUB prefill probe, not full inference. Uncovered operations are not included.

Model	Inputs	Mean / probe	Range	Coverage
ResNet	32	8.175	8.175–8.175	Counted specialist operations
GLSim	32	70.281	70.281–70.281	Counted specialist operations
GroundingDINO	32	668.259	499.213–820.176	Spatial sampling uncovered
InstanceVG	–	–	–	Not measured
BEiT3	32	76.799	73.587–83.260	Counted specialist operations
ViLT	32	84.896	57.612–120.691	Counted specialist operations
YOLO26x	32	143.222	116.224–193.928	Counted specialist operations
RT-DETR-X	32	224.644	224.644–224.644	Spatial sampling uncovered
MiniCPM	1	33,243.606	–	CUB prefill only
Qwen3.8	1	249,018.237	–	CUB prefill only

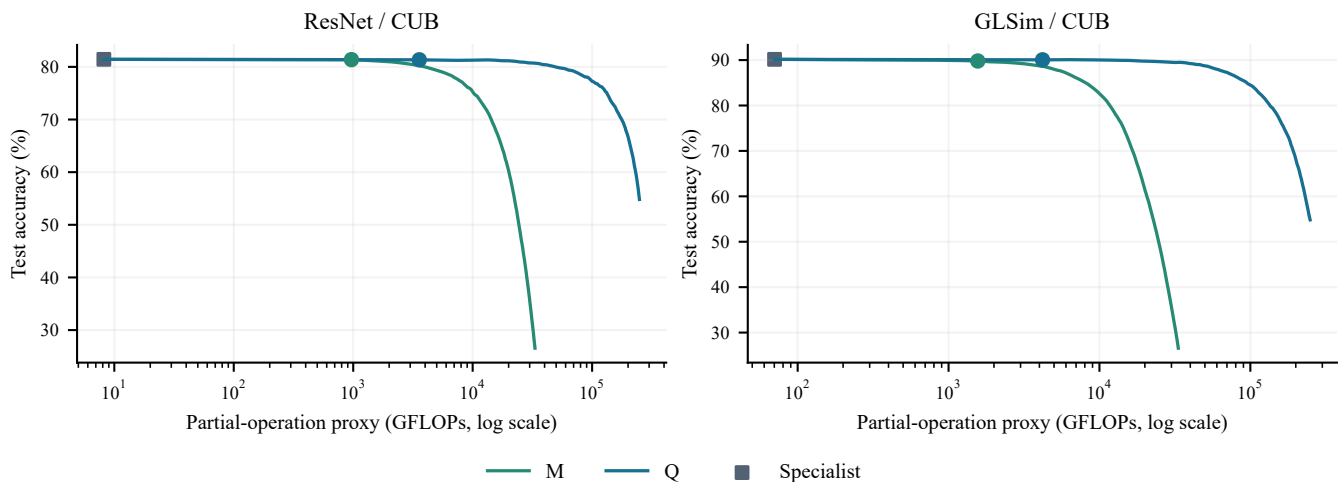


Figure S11: Illustrative CUB proxy on a logarithmic GFLOPs axis: counted mean specialist cost plus actual call fraction times a fixed-input VLM prefill probe. It omits router cost, decoding and uncovered operators, and ignores variation in VLM input cost. It is neither measured test-set computation nor a measured lower bound.

J Evidence Files, Reproducibility and Release Scope

The supplied source package contains the numerical inputs for the main tables and curves, the compact-figure and table builders, and this independent supplement. The main sixteen-pair records are in `data/results.json`; `data/budget_diagnostics.json` contains the exact-budget diagnostic. `data/earlier_results.json` and the two joint-training comparison files retain the additional studies’ original configurations and reported metrics. `data/operation_measurements.json` records the partial cost probes. The readable full-table values are regenerated from raw fractions rather than copied from rounded manuscript cells.

All results inherit the recorded task prompts, specialist thresholds, invalid-output handling and metric definitions. Student distillation excludes validation and test inputs, but cannot undo prior exposure in reused specialists. CUB specialist provenance and earlier test inspection limit claims about held-out generalization. All final comparisons use one seed; no statistical significance is asserted.

The accompanying code repository contains the router and distillation implementations, input-state pooling, task prompts, final training configurations, saved result summaries and this supplement. It also includes separate unrun configurations for final-setting input removals. Large datasets, cached features and model weights are kept outside Git and are described by the required directory layout. Redistribution follows the source datasets’ and models’ terms. Repository preparation does not imply that these external artifacts have already been publicly hosted.

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